Ch. 5 Optional – Solns

Principles of Microeconomics

September 22, 2025

1. (a) $\eta = \left|\frac{-12\%}{10\%}\right| = 1.2$

   (b)
   $$0.5 = \left|\frac{\%\Delta Q_D}{-20\%}\right|$$
   $$10\% = \%\Delta Q_D$$

   Note: $\%\Delta Q_D$ is positive because of the law of demand. If price decreases, quantity demanded increases.

   (c)
   $$1 = \left|\frac{-15\%}{\%\Delta P}\right|$$
   $$\%\Delta P = 15\%$$

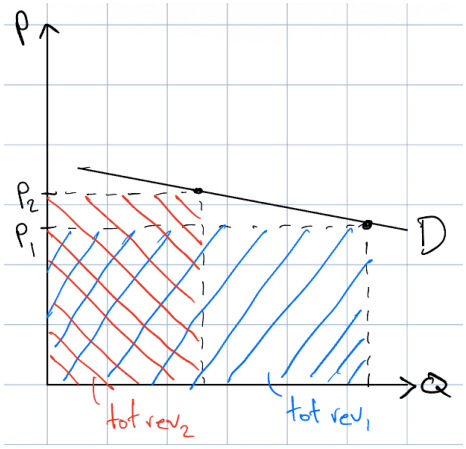
   Note: $\%\Delta P$ is positive because of the law of demand. If quantity demanded decreases, that means price must have increased.

   (d)
   $$1.2 = \frac{\%\Delta Q_S}{10\%}$$
   $$12\% = \%\Delta Q_S$$

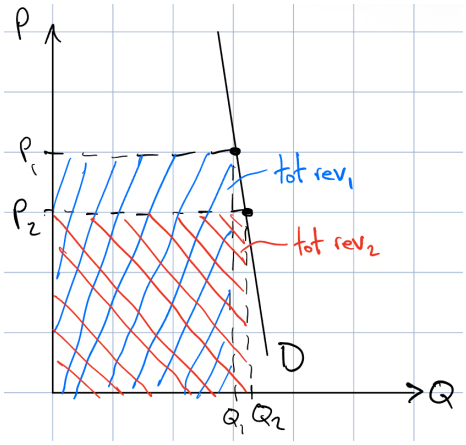
   Note: $\%\Delta Q_S$ is positive because of the law of supply. If price increases, quantity supplied increases.

   (e) $P.E.S = \frac{-10\%}{-15\%} = 0.67$

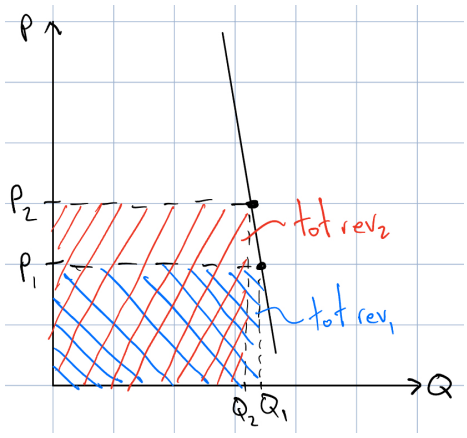
2. (a) I would not. Since demand is elastic, an increase in price will decrease quantity significantly. Quantity demanded will drop proportionately more than price will increase, so total revenue will decrease.



(b) I would not. Since demand is inelastic, a decrease in price will not increase quantity significantly. Quantity demanded will increase proportionately less than price will decrease, so total revenue will decrease.



3. (a) I disagree. Since demand is inelastic, an increase in price won't decrease quantity significantly. The increase in price will be proportionately larger than the decrease in quantity, so total revenue will go up. That means buyers will spend more on cocaine.



(b) I disagree. Since demand is elastic, a decrease in price will increase quantity significantly. The decrease in price will be proportionately smaller than the increase in quantity so total revenue will go up.

